

Representing Relationships as Functions

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Function

A **function** is a relationship in which each input corresponds to exactly one output.

Independent Variable

The **independent variable** is the variable that is chosen or controlled.

Dependent Variable

The **dependent variable** is the variable whose value depends on the independent variable.

Writing a Function

When writing a function from a situation:

Identify the independent variable.

Identify the dependent variable.

Determine how the dependent variable changes as the independent variable changes.

Write an equation that describes the relationship.

Examples

Example 1

A taxi charges a fixed flag-down rate of ₱40 plus ₱12 for every kilometer traveled.

Let x represent the number of kilometers traveled and y represent the total fare.

$$y = 40 + 12x$$

Example 2

A student earns ₱25 for every notebook sold.
Let x represent the number of notebooks sold
and y represent the total earnings.

$$y = 25x$$

Example 3

A delivery rider receives a fixed allowance of ₱100 plus ₱15 for every package delivered.

Let x represent the number of packages delivered and y represent the total amount earned.

$$y = 100 + 15x$$

Example 4

A candle is 25 cm tall and burns at a rate of 2 cm per hour.

Let x represent the number of hours and y represent the height of the candle.

$$y = 25 - 2x$$

Practice Exercises

Practice 1

A taxi charges a fixed flag-down rate of ₱40 plus ₱12 for every kilometer traveled.

Let x represent the number of kilometers traveled and y represent the total fare.

Write a **function equation**.

Practice 2

A student earns ₱25 for every notebook sold.
Let x represent the number of notebooks sold
and y represent the total earnings.
Write a **function equation**.

Practice 3

A delivery rider receives a fixed allowance of ₱100 plus ₱15 for every package delivered.

Let x represent the number of packages delivered and y represent the total amount earned.

Write a **function equation**.

Practice 4

A candle is 25 cm tall and burns at a rate of 2 cm per hour.

Let x represent the number of hours and y represent the height of the candle.

Write a **function equation**.

Practice 5

A plant is initially 12 cm tall and grows 4 cm every week.

Let x represent the number of weeks and y represent the height of the plant.

Write a **function equation**.

Practice 6

A school club collects ₱500 as an initial fund and receives ₱75 from each member who joins.

Let x represent the number of members and y represent the total fund.

Write a **function equation**.

Practice 7

A printing shop charges ₱80 as a setup fee and ₱3 per page printed.

Let x represent the number of pages printed and y represent the total cost.

Write a **function equation**.

Activities

Activity 1

A student earns ₱30 for every pen sold.

Let x represent the number of pens sold and y represent the total earnings.

Write a **function equation**.

Activity 2

A water tank already contains 15 liters of water. It is filled at a rate of 4 liters per minute. Let x represent the number of minutes and y represent the amount of water in liters.

Write a **function equation**.

Activity 3

A parking lot charges ₱40 for the first hour and ₱25 for each additional hour.

Let x represent the number of additional hours parked and y represent the total parking fee.

Write a **function equation**.

Activity 4

A candle is 30 cm tall and burns at a rate of 3 cm per hour.

Let x represent the number of hours and y represent the height of the candle.

Write a **function equation**.

Activity 5

A plant is initially 15 cm tall and grows 5 cm every week.

Let x represent the number of weeks and y represent the height of the plant.

Write a **function equation**.

Activity 6

A school club collects ₱600 as an initial fund and receives ₱80 from each member who joins.

Let x represent the number of members and y represent the total fund.

Write a **function equation**.

Thank you!